

LM119/LM219/LM319 High-Speed Dual Comparator

General Description

The LM119 series are precision high-speed dual comparators fabricated on a single monolithic chip. They are designed to operate over a wide range of supply voltages down to a single 5V logic supply and ground. Further, they have higher gain and lower input currents than devices like the LM710. The uncommitted collector of the output stage makes the LM119 compatible with RTL, DTL and TTL as well as capable of driving lamps and relays at currents up to 25 mA.

The LM319A offers improved precision over the standard LM319, with tighter tolerances on offset voltage, offset current and voltage gain.

Although designed primarily for applications requiring operations from digital logic supplies, the LM119 series are fully specified for power supplies up to $\pm 15V$. It features faster response than the LM111 at the expense of higher power dissipation.

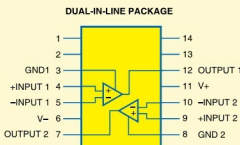
However, the high speed, wide operating voltage range and low package count make the LM119 much more versatile than older devices like the LM711.

The LM119 is specified from $-55^{\circ}C$ to $+125^{\circ}C$, the LM219 is specified from $-25^{\circ}C$ to $+85^{\circ}C$, and the LM319A and LM319 are specified from $0^{\circ}C$ to $+70^{\circ}C$.

Features

1. Two independent comparators
2. Operates from a single 5V supply
3. Typically 80ns response time at $\pm 15V$
4. Minimum fan-out of 2 each side
5. Maximum input current of 1 μA over temperature
6. Inputs and outputs can be isolated from system ground
7. High common-mode slew rate

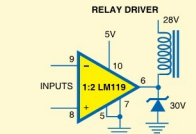
CONNECTION DIAGRAM



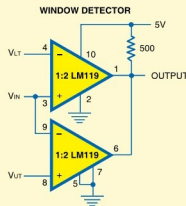
TOP VIEW
ORDER NUMBER LM119J, LM119J/883 (NOTE 1),
LM219J, LM319J, LM319AM,
LM319M, LM319AN OR LM319N
SEE NS PACKAGE NUMBER J14A, M14A OR N14A

NOTE 1: Also available per SMD# 8601401
or JM38510/10306

TYPICAL APPLICATIONS (NOTE 2)



NOTE 2: Pin numbers are for metal can package



$V_{OUT} = 5V$ FOR $V_{LT} \leq V_{IN} \leq V_{UT}$
 $V_{OUT} = 0$ FOR $V_{IN} < V_{LT}$ OR $V_{IN} > V_{UT}$

Fig. 3: Product description page of LM119/219/319

Norton op-amps a popular choice in many general-purpose op-amp application circuits. LM3900 is a popular Norton op-amp.

In the case of instrumentation op-amps, input impedance, CMRR, and values of offsets and offset drifts approach ideal values. For example, AD524 instrumentation op-amp from Analog Devices offers a differential, common-mode input resistance of 1 giga-ohm, CMRR of 120 dB, offset

voltage of 50 μV and offset drift of 0.5 $\mu V/^{\circ}C$. These features make instrumentation op-amps particularly suitable for amplifying low-level differential signals with high common-mode content (such as those available from

front-end circuits of data acquisition systems). Another salient feature of instrumentation op-amps is that the gain is programmable with a single externally connected resistor. Also, some instrumentation op-amps offer a set of fixed gain values without the use of even the lone programmable resistor. For example, AD524 can be programmed for any gain value between 1 and 1000 with a single externally connected resistor, whereas gain values of 1, 10, 100 and 1000 are achievable without any external resistor.

Isolation op-amps, also called isolation amplifiers, are op-amps with their output stage electrically isolated from the input differential stage. Isolation impedance of 1000 giga-ohms and isolation voltage of 1000 volts are common. Isolation op-amps are mainly used in medical instrumentation where isolation between the basic sensor—which is usually in physical contact with the patient's body—and the equipment is desired.

Op-amp datasheet

The datasheet of an op-amp begins with general product description. The information in this part almost invariably includes a summary of its performance parameters, product highlights, package description and, in some cases, technological aspects of the device architecture. Fig. 3, which is nothing but a compilation of the information found on the first page of the datasheet of op-amp comparator LM119/219/319, illustrates the point.

Other sections of the datasheet include essential components like absolute maximum ratings, electrical characteristics and thermal characteristics. Electrical characteristics contain a range of performance curves that tell us about the variation of major

The first step in choosing the right op-amp for a given application is to know and appreciate the category that the desired op-amp is likely to belong to. This helps in quick identification of the device.